Kavya Sridhar

San Diego, CA | k1sridhar@ucsd.edu | +1(858)-372-3468 | in kavya-sridhar | kavyasridhar1501 | Portfolio

Master's student in Data Science with a strong foundation in Machine Learning and LLMs. Passionate about leveraging data-driven solutions to drive meaningful impact.

Education

Master of Science, Data Science, *University of California San Diego* (GPA 3.916/4.0) **2024–2026**
B.E. Computer Science and Engineering, *CEG Anna University* (CGPA 9.08/10) **2019–2023**
Relevant Coursework: Linear Algebra, Probability and Statistics, Statistical Models, Data Science Programming, Object-Oriented Programming (OOP), Algorithms for Data Science, Database Management Systems, Scalable Data Systems, Big Data Analytics, Distributed Systems, Operating Systems, Machine Learning, Deep Learning, Social Network Analysis.

Experience

Société Générale Global Solution Centre **July 2023 – August 2024**
Software Engineer *Chennai, Tamil Nadu*

- Optimized CI/CD pipelines in Palantir Foundry, reducing query execution time by **20%** and accelerating decision-making for 4 business units.
- Developed interactive dashboards in Palantir Foundry Workshop, eliminating **100%** manual data analysis time and improving stakeholders' operational efficiency by **50%**.
- Enhanced data storage and workflow efficiency for new business units leveraging Scala, Spark, Amazon S3, and Oozie.

Projects

LegalInsight, Self-RAG Legal Contract Analyzer | Self-RAG, JavaScript, Flask, FAISS **Sepetember 2025**

- Implemented a client-side legal AI system implementing Self-RAG with EigenScore-inspired hallucination detection, processing 6,858 LegalBench-RAG queries with **74.65% semantic consistency** and achieving **4.26x speedup** over manual contract review across multiple LLM providers.

ChurnGuard — Banking Churn Prediction | PostgreSQL, LightGBM, React, FastAPI, MLFlow **December 2024**

- Developed an end-to-end MLOps platform with React + FastAPI + PostgreSQL stack, preprocessing 10,000+ customer records and deploying LightGBM models fine-tuned to achieve **86.5% AUC-ROC** and **77.8% precision**, with MLflow for experiment tracking and model versioning, and SHAP based explainability enabling targeted retention strategies.

Indian Sign Language (ISL) Interpreter | Tensorflow, Keras, OpenCV, **May 2022**

- Built an ISL recognition system using CNNs, achieving **98.49%** accuracy on a custom dataset, enhancing accessibility for sign language users.

Publications

D. Ramesh, I. Sriram, **K. Sridhar**, S. D. Dunston, and M. A. R. V, "***Understanding DeepFool Adversarial Attack and Defense with Skater Interpretations***" *, 2023 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET)*, Chennai, India, 2023, pp. 1-5.

- Improved the robustness of the ResNet50 architecture against adversarial attacks by **43%** through adversarial training.
- Enhanced the quantification of performance metrics by visually analyzing the impact of the attacks using Skater's visual interpretation tools.

Skills

Programming Languages: Python, C, C++
Data Engineering & Databases: MySQL, PostgreSQL, Apache Spark, Hadoop, Amazon S3, FAISS
Machine Learning & AI Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Keras
Cloud, DevOps & Tools: AWS, Docker, Git, GitHub, Jenkins, CI/CD, JIRA, MLFlow
Web Development & Frameworks: HTML, CSS, JavaScript, TypeScript, React, FastAPI, Flask
Concepts: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Data Ingestion & ETL, Database Management, MLOps